

Probabilistic Modeling of Stock Breakout Success during Consolidation Phases in the Vietnamese Equity Market

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Abstract—This paper proposes a probabilistic modeling framework to estimate the likelihood of successful stock breakouts during price consolidation phases in the Vietnamese equity market. Identifying high-probability breakout opportunities is a challenging task due to market noise, changing investor behavior, and the nonlinear dynamics of financial time series. Unlike traditional rule-based breakout strategies that rely on fixed technical thresholds, the proposed approach formulates breakout prediction as a supervised machine learning problem that outputs a calibrated probability score, enabling more robust decision-making under market uncertainty.

To capture consolidation characteristics and early breakout signals, we construct a comprehensive feature set derived from historical price action, trading volume dynamics, volatility measures, and trend-related indicators. Several machine learning algorithms are evaluated to model the complex relationship between these features and future breakout outcomes. The framework is trained and validated using historical data from a broad universe of Vietnamese listed stocks, with emphasis placed on predictive reliability, probability calibration, and model generalization rather than deterministic classification accuracy alone.

Experimental results indicate that the proposed probabilistic framework is capable of generating meaningful predictive signals for assessing breakout success. The estimated probabilities provide an interpretable measure of confidence that can support downstream applications such as signal filtering, portfolio construction, risk management, and quantitative trading strategy development in emerging equity markets.

Index Terms—Stock Breakout, Price Consolidation, Probabilistic Modeling, Machine Learning, Vietnamese Stock Market, Financial Time Series

I. INTRODUCTION

The objective of this study is to evaluate whether detected breakout events are sufficiently reliable to be considered actionable trading opportunities. In this context, breakout trading can be formulated as a signal filtering problem, where the model is designed to distinguish high-quality breakout setups from weak or false signals.

This perspective provides two key practical advantages. First, breakout strategies are highly time-efficient, as meaningful trading opportunities are typically concentrated within short periods of strong upward momentum. During these phases, price move-

ments tend to be sharp and accelerated, allowing for the potential realization of substantial returns within relatively short time windows. This enables traders to avoid prolonged market exposure during low-opportunity periods and instead focus on high-conviction events that emerge rapidly when favorable market conditions arise.

Second, the proposed framework contributes to risk reduction by naturally limiting exposure during bearish or weak market regimes, where breakout events are significantly less frequent and often lack follow-through. In such conditions, the scarcity of valid breakout signals effectively reduces unnecessary trading activity and helps avoid adverse entry timing in unfavorable market environments.

As a result, a probabilistic modeling approach enables more disciplined capital allocation by selectively prioritizing high-confidence breakout events, rather than treating all detected breakouts as equally actionable signals.

II. RELATED WORK

Breakout trading strategies constitute a fundamental component of technical analysis and quantitative trading systems. Classical breakout-based approaches are typically rule-driven and rely on predefined price and volume conditions to identify potential trend continuation. Among the most widely used frameworks are horizontal support and resistance breakouts, where a trade is triggered when price exceeds historically observed consolidation boundaries. Another prominent approach is the Donchian Channel breakout system, which defines breakout signals based on the highest high and lowest low over a fixed rolling window. In addition, volatility-adjusted breakout methods use indicators such as Average True Range (ATR) to dynamically adjust breakout thresholds according to market volatility. Volume-based confirmation strategies further require abnormal trading volume alongside price breakouts to validate signal strength.

Despite their widespread use, these methods are inherently deterministic and threshold-based. They treat all breakout events as homogeneous signals and do

not explicitly model the probability of post-breakout continuation. In practice, this leads to a high rate of false breakouts, especially in sideways or low-liquidity markets where price frequently crosses technical levels without sustained momentum. Furthermore, fixed-window methods such as Donchian channels are sensitive to parameter selection and often fail to adapt to changing market regimes, resulting in unstable performance across different market conditions.

Machine learning methods have been widely applied in financial prediction tasks, including short-term return forecasting and directional classification. Models such as gradient boosting machines, random forests, and deep neural networks are capable of capturing nonlinear relationships in financial data and integrating multiple market features. However, most existing approaches formulate the problem as generic price prediction rather than event-driven trading, and therefore do not explicitly model the quality or reliability of breakout signals.

Recent studies in financial econometrics emphasize the importance of market regimes in determining asset behavior. Empirical evidence shows that breakout effectiveness varies significantly across bullish, bearish, and high-volatility regimes. However, these regime-dependent effects are typically studied in the context of return prediction or risk modeling, and are not explicitly incorporated into breakout-specific decision frameworks.

Overall, existing literature lacks a unified probabilistic framework that explicitly models breakout success as an event-level prediction problem conditioned on market structure and regime information. In particular, there is limited work that combines consolidation structure representation, market regime awareness, and supervised learning to estimate breakout success probability. This paper addresses this gap by proposing a probabilistic model for evaluating breakout quality in the Vietnamese equity market.

III. METHODOLOGY

A. Data Pipeline

1) *Input Data*: The dataset is constructed as a cross-sectional panel consisting of approximately 255 Vietnamese equities with market capitalization greater than 2,500 billion VND, ensuring sufficient liquidity and tradability. The selected universe focuses on mid- and large-cap stocks across major sectors such as banking, real estate, energy, and industrial manufacturing.

For each stock i and time t , we collect daily OHLCV data:

$$X_{i,t} = \{Open_{i,t}, High_{i,t}, Low_{i,t}, Close_{i,t}, Volume_{i,t}\} \quad (1)$$

All prices are adjusted values to account for corporate actions, ensuring consistency across time.

In addition to stock-level data, we collect VNINDEX market index data from VCI source over the period 2009-06-01 to 2026-12-31, defined as:

$$M_t = \{Open_t, High_t, Low_t, Close_t, Volume_t\} \quad (2)$$

This provides a synchronized market benchmark aligned with individual stock time series.

2) *Preprocessing*: All raw datasets undergo strict preprocessing steps prior to feature engineering:

- Conversion of all date fields to a unified datetime format (*Ngay*)
- Removal of missing or invalid timestamps
- Sorting stock data by $(stock_id, t)$ and market data by (t)
- Standardization of column names across datasets

To ensure strict causal integrity, all future leakage is eliminated by enforcing temporal alignment constraints. Specifically, all subsequent feature construction (performed in later stages) will be based only on information available at time $t - 1$:

$$X_t \leftarrow X_{t-1} \quad (3)$$

No normalization or transformation is applied at this stage in order to preserve raw market structure for downstream feature engineering.

3) *Data Universe Definition*: The trading universe is restricted to approximately 255 liquid Vietnamese equities with market capitalization exceeding 2,500 billion VND. This filtering ensures:

- sufficient liquidity for realistic execution
- reduced microcap noise and manipulation effects
- stability of statistical properties across time

The full dataset is further restricted to the period:

$$t \geq 2010-01-01 \quad (4)$$

to ensure sufficient historical depth for both stock-level dynamics and subsequent event modeling.

4) *Data Merging*: The final panel dataset is constructed by aligning stock-level observations with market index data on the time index t :

$$D_{i,t} = X_{i,t} \cup M_t \quad (5)$$

This is implemented as a time-based left join:

$$(stock_id, t) \rightarrow (stock_id, t, M_t) \quad (6)$$

After merging, the dataset is sorted by $(stock_id, t)$ to preserve temporal ordering, forming a unified panel structure that serves as the input for subsequent breakout event detection and labeling modules.

B. Data Processing Pipeline

We construct a strict causal time-series pipeline using Vietnamese stock market data from 2010 onwards, consisting of 255 large-cap stocks (market capitalization above 2,500 billion VND). All transformations are designed to avoid look-ahead bias.

1) *Market Feature Construction*: From the VNIN-DEX index, we construct a set of market regime features including:

- **Returns**: 1-day, 20-day, and 60-day percentage returns.
- **Trend Features**: Moving averages (MA50, MA200), price distance from moving averages, and MA slopes over a 20-day window.
- **Volatility Features**: Annualized rolling volatility over 20-day and 60-day windows, as well as volatility ratios between short and long horizons.
- **Risk Measures**: ATR-based volatility, return z-scores over a 20-day window, and 1-year draw-down.
- **Liquidity Proxy**: Rolling average volume and relative volume compared to historical averages.

All market features are shifted by one trading day to ensure strict causality.

2) *Market Regime Labeling*: The market state is categorized into three regimes:

- **Downtrend (0)**: Characterized by deep draw-downs and negative MA200 slope or sustained weakness below MA200.
- **Neutral (1)**: Intermediate market conditions.
- **Uptrend (2)**: Defined by shallow drawdowns, positive MA200 slope, and relatively stable volatility conditions.

3) *Feature Merging*: Market features are merged into individual stock-level data using the trading date as the key. This results in each stock observation being augmented with contemporaneous market context information.

4) *Breakout Event Construction*: We define breakout events at the stock level based on the following conditions:

- Price breaks above a rolling high over a 60–100 day lookback window.
- Confirmation through closing price above breakout threshold.
- Volatility (ATR) below a predefined threshold.
- Volume confirmation when available.

For each breakout event, a preceding base period (12–120 days) is analyzed to extract structural features such as:

- Price compression (volatility contraction),
- Base depth and tightness,
- Volume dry-up patterns,
- Pre-breakout stability of returns.

5) *Label Construction*: Each breakout event is treated as a simulated trade:

- **Entry price**: next-day open after breakout.
- **Take-profit (TP)**: $\text{entry} + k \times \text{ATR}$.
- **Stop-loss (SL)**: $\text{entry} - k \times \text{ATR}$.

Within a fixed lookahead window (up to 120 trading days), the trade outcome is determined as:

- Label = 1 if TP is hit first.

- Label = 0 if SL is hit first.
- Otherwise, a threshold-based rule is applied using maximum realized return.

Additional metrics are recorded for each trade, including Maximum Favorable Excursion (MFE), Maximum Adverse Excursion (MAE), realized return, and time-to-exit statistics.

6) *Final Dataset Filtering*: The final dataset is filtered to include only observations from 2010 onwards after label construction, ensuring consistency and validity of all engineered features and labels.

C. SEPA Feature Engineering and Training Dataset Construction

1) *SEPA Feature Engineering*: We construct a structured feature set inspired by the SEPA-style breakout framework to capture key aspects of trend, volatility, volume, and resistance pressure.

SEPA (Specific Entry Point Analysis), originally popularized by Mark Minervini, is a momentum investing approach focused on identifying high-probability entry points during early breakouts from well-formed consolidation bases.

In this study, we adapt SEPA into a systematic, data-driven framework suitable for machine learning. Instead of using it as a discretionary trading methodology, we reformulate it into an event-driven representation learning problem, where each sample corresponds to a candidate breakout event.

This transformation converts SEPA into a structured feature engineering pipeline that enables the model to learn statistical patterns associated with successful breakout setups, including consolidation structure, volatility compression, liquidity expansion, and trend alignment across multiple time horizons.

This design preserves the core economic intuition of SEPA—focusing on high-quality setups—while enabling scalable, consistent, and empirical evaluation through machine learning models.

a) *Trend Alignment Features*: We compute multi-horizon moving averages at the stock level:

- MA50, MA150, MA200
- MA ratio relationships: MA50/MA150, MA150/MA200
- Price distance from MA50

These features measure medium- and long-term trend alignment.

b) *52-week Relative Strength*: We define:

- Distance from 52-week high
- 60-day return
- Cross-sectional RS percentile ranking by date
- Relative Strength new high indicator (RS line vs 120-day high)

where RS line is defined as:

$$RS = \frac{P_{stock}}{P_{market}} \quad (7)$$

c) *Volatility and Base Structure*: We construct volatility and base formation features:

- True Range (TR) and ATR(14), ATR(60)
- Volatility compression ratio: $\text{ATR}(14)/\text{ATR}(60)$
- Base depth measured by 60-day high-low range
- Price tightness measured by 20-day rolling standard deviation

d) *Volume Features*:

- Volume MA20 and MA50
- Breakout volume ratio (current volume / MA20)

e) *Resistance Pressure Features*: We define:

- Rolling 30-day resistance level
- Near-resistance indicator (within 3% of resistance)
- Resistance test frequency over 30-day window

2) *Hard Filtering Strategy*: We apply a filtering step to ensure high-quality breakout candidates:

- Distance from 52-week high > -0.35
- RS percentile (60D) > 0.5
- Base depth > -0.4
- Volatility compression ratio < 0.95

This step reduces noise and improves signal quality before model training.

3) *Dataset Construction*: We define training samples only at breakout events where:

- $is_breakout = 1$
- label is available (trade outcome)

After filtering, the dataset contains:

- 1,166 total samples
- Balanced classes:
 - Class 0: 566 (48.5%)
 - Class 1: 600 (51.5%)

4) *Train-Test Split*: We split data using a temporal cutoff:

- Train: before 2024-01-01
- Test: from 2024-01-01 onwards

Result:

- Train samples: 950
- Test samples: 216
- Stable winrate distribution across splits

5) *Model Training and Cross-Validation*: To ensure temporal consistency and avoid look-ahead bias, we perform model validation strictly on pre-2024 data using a time-series cross-validation scheme.

We adopt a LightGBM-based classification model due to its strong performance on tabular financial data and its ability to capture nonlinear interactions between engineered features.

a) *Class Imbalance Handling*: Since the dataset exhibits slight class imbalance, we compute a dynamic weighting factor:

- positive class ratio: $p = \text{mean}(y_{train})$
- $\text{scale_pos_weight} = (1 - p)/p$

This weighting is incorporated into the loss function to stabilize training and reduce bias toward the majority class.

b) *Model Configuration*: We use a gradient boosting tree model with the following hyperparameters:

- $n_estimators = 2000$
- $learning_rate = 0.02$
- $num_leaves = 24$
- $min_child_samples = 15$
- $subsample = 0.8$
- $colsample_bytree = 0.8$
- $reg_alpha = 0.5$
- $reg_lambda = 1.0$

Early stopping (50 rounds) is applied based on validation AUC to prevent overfitting.

c) *Time-Series Cross-Validation*: We apply TimeSeriesSplit with 3 folds, ensuring chronological consistency where training data always precedes validation data. Each fold simulates a realistic trading scenario where models are evaluated only on future unseen data relative to the training window.

d) *Cross-Validation Results*: The model achieves the following performance on pre-2024 validation data:

- Mean CV AUC: 0.5736
- Standard deviation: 0.0361

These results indicate that the model captures a weak-to-moderate discriminative signal in distinguishing between successful and unsuccessful breakout events. The relatively modest AUC reflects the inherent difficulty of short-horizon financial prediction under strong non-stationarity, strict causal constraints, and event-based sampling conditions.

e) *Out-of-Sample Performance*: After training on the full pre-2024 dataset, we evaluate the model on unseen post-2024 data:

- Out-of-sample AUC (2024+): 0.6322

The improvement in out-of-sample performance compared to cross-validation suggests improved generalization under the given temporal split. The AUC is interpreted strictly as a metric of ranking and discrimination ability between breakout outcomes, rather than as an indicator of trading profitability or economic performance.

f) *Economic Interpretation*: Although AUC is a classification metric, in this framework it serves as a proxy for ranking quality of trading opportunities rather than direct prediction accuracy. A higher AUC implies better separation between successful and failed breakout setups, which can be translated into improved signal selection in a portfolio construction setting.

In practice, this corresponds to improving the hit-rate of selected trades rather than predicting exact price movements. Therefore, the model is more appropriately interpreted as a *trade ranking model* rather than a pure direction forecasting system.

g) *Limitations*: Despite promising results, several limitations remain:

- **Label construction bias**: The use of fixed ATR-based take-profit and stop-loss introduces rule-

based assumptions into the ground truth definition.

- **Metric mismatch:** The evaluation relies on AUC, which does not directly reflect profitability after transaction costs, slippage, or execution constraints.
- **Market scope:** The study is limited to a single emerging market (Vietnam), which may reduce generalizability to other market structures.
- **Event selection dependency:** The model is conditioned on predefined breakout detection rules, which may exclude alternative profitable setups.

h) Conclusion: This study proposes a structured event-driven machine learning framework for breakout prediction in the Vietnamese equity market. By reformulating the SEPA-inspired discretionary trading philosophy into a systematic feature engineering and labeling pipeline, we construct a dataset that explicitly models pre-breakout market structure, including trend alignment, volatility compression, liquidity dynamics, and resistance pressure.

Empirical results demonstrate that the proposed framework achieves a mean cross-validation AUC of 0.5736 on pre-2024 data and an out-of-sample AUC of 0.6322 on unseen post-2024 data. The improvement in out-of-sample performance suggests that the model is able to capture robust structural patterns that generalize across different market regimes, despite the inherent non-stationarity of financial time series.

Rather than functioning as a pure directional forecasting system, the proposed approach is better interpreted as a trade opportunity ranking model, where predictive power is derived from identifying high-quality breakout setups rather than predicting continuous price movements.

Future work will focus on integrating portfolio-level evaluation metrics, incorporating transaction cost modeling, and extending the framework to multi-asset ranking and reinforcement learning-based execution systems.

IV. EXPERIMENTS

A. Evaluation Setup

We evaluate the proposed event-driven breakout framework on an out-of-sample test set from 2024 onward, consisting of 216 strictly time-separated breakout events. Each event is assigned a probability score from the trained LightGBM model, and performance is assessed via ranking of predicted probabilities against realized trade outcomes.

B. Baseline and Ranking Performance

The baseline win rate of the test set is 0.4907, indicating a near-balanced distribution between successful and failed breakouts.

a) Top-K and Quantile Analysis: The model shows strong concentration of successful trades in top-ranked predictions:

- Top 3: 66.67% win rate
- Top 6: 83.33% win rate
- Top 10: 90.00% win rate
- Top 5%: 90.0%, Top 10%: 80.95%, Top 20%: 74.42%

b) Precision@K:

- Precision@3 = 0.6667
- Precision@6 = 0.8333
- Precision@10 = 0.9000
- Precision@20 = 0.8000

These results confirm strong and stable ranking capability across different selection thresholds.

C. Regime-Conditioned Performance

Under a probability threshold of 0.8, the model remains robust across market regimes:

- Neutral regime: 29 trades, win rate = 75.86%
- Uptrend regime: 11 trades, win rate = 72.73%

This suggests consistent performance with slightly stronger signals in neutral market conditions.

D. Trading Simulation Metrics

We simulate a simple threshold strategy (probability > 0.8):

- Number of trades: 40
- Win rate: 72.5%
- Total return: 5.58
- Average return per trade: 13.94%
- Profit factor: 4.89

These results indicate strong asymmetric payoff characteristics, though they exclude transaction costs, slippage, and liquidity effects, and should be interpreted as signal-quality evaluation rather than realistic backtest performance.

E. Discussion

Overall, results show that while global classification performance is moderate, the model exhibits strong ranking ability, concentrating successful trades in the top-scored subset. This aligns with the intended use case of selecting high-quality breakout opportunities rather than predicting all outcomes.

Performance remains stable across market regimes, indicating robustness under moderate distribution shifts.

F. Summary of Key Findings

- Strong ranking ability with significant win rate concentration in top predictions
- High Precision@K across multiple selection thresholds
- Robust performance across market regimes
- Positive simulated trading performance with high profit factor (4.89)

V. CONCLUSION

In this study, we propose an event-driven breakout prediction framework that reformulates traditional time-series forecasting into a structured trade-event modeling problem. Instead of predicting price at arbitrary timestamps, the framework focuses on identifying high-quality breakout setups using market regime features, SEPA-inspired structural indicators, and strictly causal feature engineering.

The proposed representation captures key aspects of market behavior, including trend alignment, volatility compression, volume dynamics, and resistance pressure. By explicitly modeling pre-breakout structure, the framework approximates discretionary momentum trading in a systematic and data-driven manner.

Empirical results show that although overall classification performance is moderate, the model exhibits strong ranking ability. Out-of-sample evaluation demonstrates clear concentration of successful trades in top-ranked predictions, with Precision@K up to 0.90 and win rates in the top percentile exceeding 80–90%. A simple threshold-based simulation further confirms economic relevance, achieving a 72.5% win rate and a profit factor of 4.89 on unseen 2024 data.

These results indicate that the main value of the framework lies not in full-market prediction accuracy, but in effectively filtering and prioritizing asymmetric trading opportunities. The combination of event modeling and ranking-based learning provides a practical approach for translating market microstructure signals into actionable decisions.

However, limitations remain, including dataset size, regime sensitivity, and only moderate AUC performance. Future work may focus on larger universes, multi-market validation, and advanced sequence modeling methods.

Overall, the study demonstrates that event-centric learning with SEPA-inspired feature design is a promising direction for building interpretable and economically meaningful trading models.

REFERENCES

- [1] H. Markowitz, "Portfolio selection," *The Journal of Finance*, vol. 7, no. 1, pp. 77–91, 1952.
- [2] E. F. Fama, "Efficient capital markets: A review of theory and empirical work," *The Journal of Finance*, vol. 25, no. 2, pp. 383–417, 1970.
- [3] E. F. Fama and K. R. French, "Common risk factors in the returns on stocks and bonds," *Journal of Financial Economics*, vol. 33, no. 1, pp. 3–56, 1993.
- [4] M. M. Carhart, "On persistence in mutual fund performance," *The Journal of Finance*, vol. 52, no. 1, pp. 57–82, 1997.
- [5] S. J. Grossman and J. E. Stiglitz, "On the impossibility of informationally efficient markets," *American Economic Review*, vol. 70, no. 3, pp. 393–408, 1980.
- [6] A. W. Lo and A. C. MacKinlay, "Stock market prices do not follow random walks: Evidence from a simple specification test," *The Review of Financial Studies*, vol. 1, no. 1, pp. 41–66, 1988.
- [7] A. W. Lo, "The adaptive markets hypothesis: Market efficiency from an evolutionary perspective," *Journal of Portfolio Management*, vol. 30, no. 5, pp. 15–29, 2004.
- [8] S. Gu, B. Kelly, and D. Xiu, "Empirical asset pricing via machine learning," *Review of Financial Studies*, vol. 33, no. 5, pp. 2223–2273, 2020.
- [9] B. Kelly and D. Xiu, "Financial machine learning," *Foundations and Trends in Finance*, vol. 13, no. 3–4, pp. 205–363, 2023.
- [10] J. Sirignano and R. Cont, "Universal features of price formation in financial markets," *Quantitative Finance*, vol. 19, no. 9, 2019.
- [11] M. López de Prado, "Building diversified portfolios that outperform out of sample," *The Journal of Portfolio Management*, vol. 42, no. 4, 2016.
- [12] A. W. Lo, H. Mamaysky, and J. Wang, "Foundations of technical analysis: Computational algorithms, statistical inference, and empirical implementation," *The Journal of Finance*, vol. 55, no. 4, pp. 1705–1765, 2000.
- [13] M. Minervini, *Trade Like a Stock Market Wizard*, McGraw-Hill Education, 2013.
- [14] M. Minervini, *Think and Trade Like a Champion*, McGraw-Hill Education, 2015.
- [15] G. Ke et al., "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [16] T. Bergmeir and J. M. Benítez, "On the use of cross-validation for time series predictor evaluation," *Information Sciences*, vol. 191, pp. 192–213, 2012.
- [17] D. H. Bailey et al., "The probability of backtest overfitting," *Journal of Computational Finance*, 2014.